Strong Weihrauch reduction of the Schreier free-set theorem to binary Ramsey

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Editorial note. The construction425 and solution443 work with positive minimum. The exceptional singleton {0} can be colored arbitrarily and0 removed from the homogeneous solution before subtracting1.

Editorial note. Only the strong Weihrauch reduction is counted, not the later reverse-mathematical corollary or a new omega-jump lower bound.

Editorial scope. One original result. Full source-local mathematical construction included; named classical inputs are explicit.

Allowed standard prerequisites (editorial).

- Computable recursion on finite sequences and the Schreier lexicographic well-order. Exactly-large finite sets are lexicographically well-ordered; an effectively specified recursive call to a smaller tuple terminates. Basic oracle-computability and strong Weihrauch definitions are supplied.

There exist four other subsystems which, together with RCA_0, calibrate the strength of most theorems. These subsystems are known as the "Big Five" (see Montálban [2]). Among these, ACA_0 (Arithmetic Comprehension Axiom) extends the axioms of RCA_0 with the comprehension scheme for all arithmetic formulas. Based on the correspondence between computability and definability, it is equivalent to stating the existence of the Turing jump of any set ($\forall X \exists Y (Y = X')$).

We shall also consider two lesser-known stronger variants of ACA_0, namely, ACA_0' and ACA_0^+. The system ACA_0' extends RCA_0 with the axiom stating closure under all finite jumps; the system ACA_0^+ extends RCA_0 with the axiom stating the existence of the ω-jump of any set. The ω-jump of a set X is the set $X^{(\omega)} = \bigoplus_n X^{(n)}$. The system ACA_0^+ is famous for being the best known upper bound to Hindman's theorem.

Let us first fix some notation and recall the definition of these principles for colorings of finite subsets of fixed size. We denote by $\mathbb{N}$ the set of natural numbers and by $\mathbb{N}^+$ the set of positive natural numbers. For $X \subseteq \mathbb{N}$ and $n \in \mathbb{N}$ we denote by $[X]^n$ the set of all n-subsets of X. We denote by $[X]^{<\omega}$ the set of all finite subsets of X. We always assume that sets are presented in increasing order and make no distinction between a set and the sequence of its elements in increasing order. We write $x \cdot y$ for the concatenation of two sequences x and y. We identify a natural number with the set of its predecessors, so that, if $k \geq 1$ we can write $f : X \to k$ to declare a function from X to $\{0, 1, \ldots, k-1\}$. A function of this type is often called a coloring of X in k colors.

Definition 0.1. Q *is* strongly Weihrauch reducible *to* P *(denoted* $\mathsf{Q} \leq_{\mathrm{sW}} \mathsf{P}$*) if there exist Turing functionals* Φ *and* Ψ *such that for every instance* X *of* Q *we have that* $\Phi(X)$ *is an instance of* P*, and if* Y *is a solution to* P *for* $\Phi(X)$ *then* $\Psi(Y)$ *is a solution to* Q *for* X*.*

Definition 0.2 ((Exactly) ω-large sets)**.** *A finite* $s \subseteq \mathbb{N}$ *is* ω*-large if* $|s| \geq 1 + \min s$ *and is exactly* ω*-large if* $|s| = 1 + \min s$*.*

The ω-large sets are also known as *relatively large* sets in the literature. For an infinite $X \subseteq \mathbb{N}$ we denote by $[X]^{!\omega}$ the family of all exactly ω-large subsets of X.

Definition 0.3 (Large Ramsey Theorem)**.** *Let* $k \in \mathbb{N}^+$*. For every coloring* $f : [\mathbb{N}]^{!\omega} \to k$*, there exists an infinite set* $H \subseteq \mathbb{N}$ *such that* $|f([H]^{!\omega})| = 1$*. We abbreviate this statement by* $\mathsf{RT}^{!\omega}_k$*.*

Definition 0.4 (Large Free Set Theorem)**.** *For every coloring* $f : [\mathbb{N}]^{!\omega} \to \mathbb{N}$ *there exists an infinite set* $H \subseteq \mathbb{N}$ *such that for every set* $s \in [H]^{!\omega}$*,* $f(s) \notin (H \backslash s)$*. The set* H *is called* free *for* f*. We abbreviate this statement by* $\mathsf{FS}^{!\omega}$*.*

Theorem 0.5. $\mathsf{FS}^{!\omega} \leq_{\mathrm{sW}} \mathsf{RT}^{!\omega}_2$.

Proof. For $s = \{s_0, \dots, s_{s_0}\}$ an exactly ω-large set with $s_0 > 0$, and $s_0 < \cdots < s_{s_0}$ define $s \ominus 1$ to be the following exactly ω-large set: $\{s_0 - 1, s_1 - 1, \dots, s_{s_0 - 1} - 1\}$. The idea is to keep one degree of freedom: s_{s_0}, as in the proof of Carlucci and Zdanowski [1, Proposition 4.1].

Let $f : [\mathbb{N}]^{!\omega} \to \mathbb{N}$ be an instance of $\mathsf{FS}^{!\omega}$. Consider the following function $g : [\mathbb{N}]^{!\omega} \to 2$ defined by induction. Note that the function g is called recursively on lexicographically smaller parameters, so the induction is well-defined since the lexicographic order is a well-order. For $s = \{s_0, \dots, s_{s_0}\}$ an exactly ω-large set:

$$g(s) = \begin{cases} 0 & \text{if } f(s \ominus 1) = s_i - 1 \text{ for some } i < s_0 \\ 1 - g(f(s \ominus 1) + 1, s_1, \dots, s_{f(s \ominus 1)+1}) & \\ & \text{if } f(s \ominus 1) < s_0 - 1 \\ 1 - g(s_0, \dots, s_i, f(s \ominus 1) + 1, s_{i+2}, \dots, s_{s_0}) & \\ & \text{if } f(s \ominus 1) \in (s_i - 1, s_{i+1} - 1) \text{ for some } i < s_0 - 1 \\ 0 & \text{if } f(s \ominus 1) \in (s_{s_0 - 1} - 1, s_{s_0} - 1) \\ 1 & \text{otherwise } (\textit{if } f(s \ominus 1) \geq s_{s_0}) \end{cases}$$

Let $H = \{x_0, x_1, \dots\} \subseteq \mathbb{N}^+$ be an infinite g-homogeneous set as given by $\mathsf{RT}^{!\omega}_2$. We claim that $H' = \{x_0 - 1, x_1 - 1, \dots\}$ is f-free. Consider some exactly ω-large set $\{s_0 - 1, \dots, s_{s_0 - 1} - 1\} \subseteq H'$. Then $\{s_0, \dots, s_{s_0 - 1}\} \subseteq H$. There are two cases:

Case 1: H is homogeneous for the color 0. Then, take s_{s_0} to be the next element of H after $s_{s_0 - 1}$ and write $s = \{s_0, \dots, s_{s_0}\}$, then $g(s) = 0$. There are four subcases:

Subcase 1.1: $f(s \ominus 1) = s_i - 1$ for some $i < s_0$. In that case we are done.

Subcase 1.2: $f(s \ominus 1) \in (s_{s_0 - 1} - 1, s_{s_0} - 1)$. In that case, by definition of s_{s_0}, $f(s \ominus 1)$ is not in H'.

Subcase 1.3: $f(s \ominus 1) < s_0 - 1$ and $g(f(s \ominus 1) + 1, s_1, \ldots, s_{f(s\ominus 1)+1}) = 1$. If $f(s \ominus 1) \in H'$ this contradicts the fact that H is g-homogeneous for the color 0.

Subcase 1.4: $f(s \ominus 1) \in (s_i - 1, s_{i+1} - 1)$ for some $i < s_0 - 1$ and $g(s_0, \ldots, s_i, f(s \ominus 1) + 1, s_{i+2}, \ldots, s_{s_0}) = 1$. If $f(s \ominus 1) \in H'$ this contradicts the fact that H is g-homogeneous for the color 0.

Case 2: H is homogeneous for the color 1. Take $s_{s_0} \in H$ bigger than s_{s_0-1} and write $s = \{s_0, \ldots, s_{s_0}\}$, then $g(s) = 1$. There are three subcases:

Subcase 2.1: $f(s \ominus 1) \geq s_{s_0}$. This case is impossible, indeed, as H is infinite, there exists some element $x \in H$ such that $x > f(s \ominus 1) + 1$ and therefore $f(s \ominus 1) \in (s_{s_0} - 1, x - 1)$, which leads to $g(s_0, \ldots, s_{s_0-1}, x) = 0$ contradicting the fact that H is g-homogeneous for the color 1.

Subcase 2.2: $f(s \ominus 1) < s_0 - 1$ and $g(f(s \ominus 1) + 1, s_1, \ldots, s_{f(s\ominus 1)+1}) = 0$. If $f(s \ominus 1) \in H'$ this contradicts the fact that H is g-homogeneous for the color 1.

Subcase 2.3: $f(s \ominus 1) \in (s_i - 1, s_{i+1} - 1)$ for some $i < s_0 - 1$ and $g(s_0, \ldots, s_i, f(s \ominus 1) + 1, s_{i+2}, \ldots, s_{s_0}) = 0$. If $f(s \ominus 1) \in H'$ this contradicts the fact that H is g-homogeneous for the color 1.

Notice that g is uniformly computable in f and that H' is uniformly computable in H. Therefore, $\mathsf{FS}^{!\omega} \leq_{\mathrm{sW}} \mathsf{RT}^{!\omega}_2$.

□

References

[1] Lorenzo Carlucci and Konrad Zdanowski. The strength of Ramsey's theorem for coloring relatively large sets. *J. Symb. Log.*, 79(1):89–102, 2014.

[2] Antonio Montalbán. Open questions in reverse mathematics. *Bulletin of Symbolic Logic*, 17(03):431–454, 2011. Publisher: Cambridge Univ Press.